



AMENDMENT NO. 3 MAY 1982
TO
IS : 8496-1977 SPECIFICATION FOR
PROPOXUR, TECHNICAL

Alteration

[*Pages 5 and 6, Appendix A* { see also Amendment No. 1 }]— Substitute the following for the existing Appendix:

APPENDIX A
[*Table 1, Item (i)*]

DETERMINATION OF PROPOXUR CONTENT

A-O. GENERAL

A-0.1 Either of the two methods, namely ultra-violet (uv) spectrophotometric method (see **A-1**) or saponification method (see **A-2**) may be used for the determination of propoxur content. However, ultra-violet spectrophotometric method (see **A-1**) shall be the referee method in case of dispute.

A-1. ULTRA VIOLET SPECTROPHOTOMETRIC METHOD

A-1.1 Principle — 2-Isopropoxy phenol liberated by alkaline hydrolysis is determined from the differential ultra-violet spectrum of its alkaline solution against acidic solution at 292 nm.

A-1.2 Apparatus

A-1.2.1 UV Spectrophotometer

A-1.2.2 Volumetric Flasks — 100 ml capacity.

A-1.2.3 Pipette — 10 ml capacity.

A-1.2.4 Cuvettes — 1 cm quartz.

A-1.3 Reagents

A-1.3.1 Ethyl Alcohol

A-1.3.2 Sodium Hydroxide Solution — 1 N.

Price Rs 1.50

A-1.3.3 Sulphuric Acid Solution — 1 N.

A-1.3.4 Propoxur — of known 99 percent plus purity.

A-1.4 Preparation of Standard and Sample Solutions — It is very important in preparing the *Standard* and *Sample Solutions* that volume fluctuations with changing temperatures are avoided. This is accomplished by bringing each solution to the volume mark just before proceeding to the next step in the procedure as indicated.

A-1.4.1 Preparation of Standard Solution

A-1.4.1.1 Accurately weigh 0.37 to 0.38 g of pure propoxur (see A-1.3.4) in a 100 ml volumetric flask. Add ethyl alcohol to dissolve and make up the volume with alcohol. Call this as *Standard Solution A*.

A-1.4.1.2 Transfer 10 ml of *Standard Solution A* into another 100 ml volumetric flask. Add 50 ml of ethyl alcohol and 2.5 ml of 1 N sodium hydroxide solution. Keep the mixture at room temperature for 1 hour. Dilute to mark with alcohol and shake well to get uniform solution. Call this as *Standard Solution B*.

A-1.4.2 Preparation of Sample Solutions

A-1.4.2.1 Accurately weigh a quantity of the sample containing 0.350 ± 0.020 g of propoxur in 100 ml volumetric flask. Add ethyl alcohol to dissolve and make up the volume. Call this as *Sample Solution A*.

A-1.4.2.2 Transfer 10 ml of *Sample Solution A* into another 100 ml volumetric flask. Add 50 ml of ethyl alcohol and 2.5 ml of 1N sodium hydroxide solution. Keep the mixture at room temperature for one hour. Dilute to mark with ethyl alcohol and shake well to get uniform solution. Call this as *Sample Solution B*.

A-1.5 Procedure

A-1.5.0 To obtain accurate results it is important that analysis of *Standard* and *Sample Solutions* be carried simultaneously, through the procedure given in **A-1.5.1**.

A-1.5.1 Pipette out 10 ml each of *Standard Solution B* (see A-1.4.1.2) and *Sample Solution B* (see A-1.4.2.2) into two separate sets of 100 ml volumetric flasks. To one set of flasks comprising *Standard* and *Sample Solutions* add 20 ml of 1 N sulphuric acid solution while to another set of flasks add 20 ml of 1 N sodium hydroxide solution. Make up the volume in each flask with ethyl alcohol. After thorough mixing take the UV transmittance/absorbance curve from 275 to 325 nm, taking acidic solution in reference cell and alkaline solution in sample cell. The maximum absorption will be at around 292 nm.

NOTE --- The absorbance measurement of the *Standard* and *Sample Solutions* shall agree within ± 10 percent of each other.

A-1.6 Calculation

A-1.6.1 Propoxur content, percent
by mass
$$= \frac{A_2}{A_1} \times \frac{M_1}{M_2} \times P$$

where

A_1 = absorbance of standard solution;

A_2 = absorbance of sample solution;

M_1 = mass, in g, of pure propoxur;

M_2 = mass, in g, of sample taken for test; and

P = percent purity of pure propoxur.

A-2. SAPONIFICATION METHOD

A-2.1 Principle — Propoxur is saponified with potassium hydroxide in diethylene glycol and the methyl amine formed is directly absorbed in boric acid solution and titrated against standard hydrochloric acid.

A-2.2 Apparatus — Assemble the apparatus as shown in Fig. 1.

A.2.3 Reagents

A-2.3.1 Potassium Hydroxide in Diethylene Glycol — 2 N. Dissolve 56 g of potassium hydroxide in a minimum quantity of water and make up to 500 ml with diethylene glycol.

A-2.3.2 Boric Acid Solution — 2 percent (*m/v*). Prepare by dissolving 20 g boric acid in about 500 ml of hot water, cool, add 2 ml of 0.1 percent bromocresol green, dilute to one litre, and mix. This solution need not be prepared accurately and may be prepared in a large volume sufficient for several weeks' use. If the colour fades on standing, more indicator may be added to the remaining portion.

A-2.3.3 Nitrogen

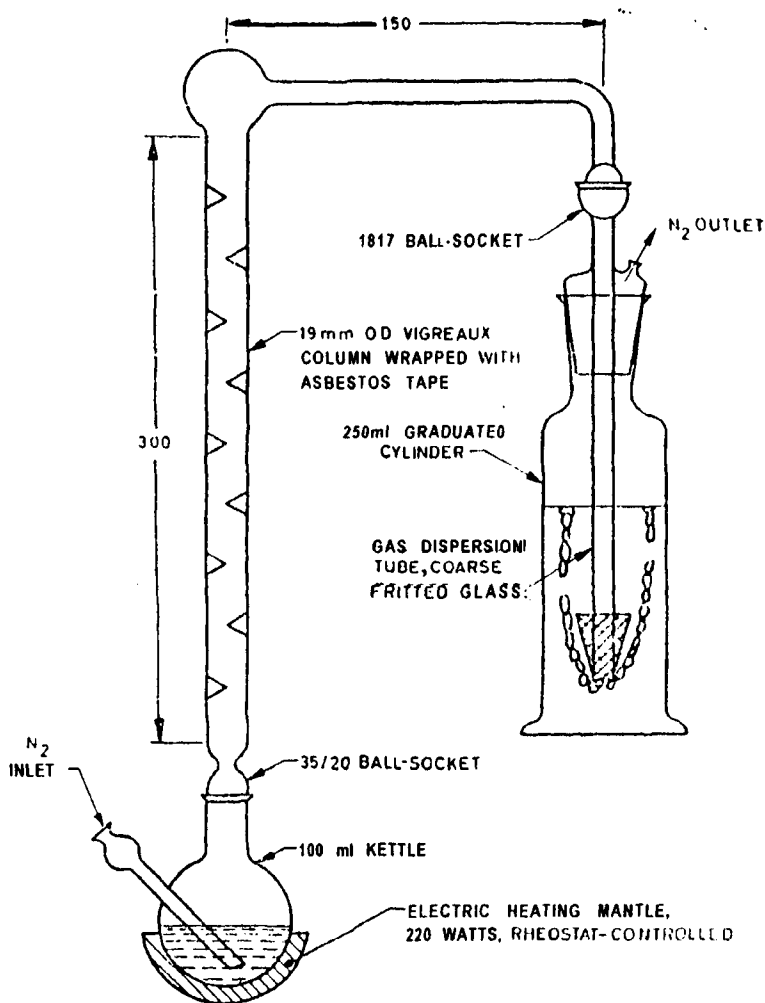
A-2.3.4 Bromocresol Green Indicator Solution — 1 percent (*m/v*). Dissolve bromocresol green in ethyl alcohol alternatively prepare an aqueous solution of the sodium salt as follows.

A-2.3.4.1 Rub 100 mg of bromocresol green in an agate mortar with 2.9 ml of 0.5 N sodium hydroxide. After the indicator has dissolved completely, add sufficient water to make the volume up to 100 ml.

A-2.3.5 Hydrochloric Acid — 0.1 N.

A-2.4 Procedure — Weigh accurately about 0.5 g of propoxur in the 100-ml reaction flask. Add 50 ml of potassium hydroxide in diethylene glycol and connect the flask to the evolution assembly and nitrogen. Place 100 ml of boric acid solution in the 250-ml measuring cylinder and connect the same to the gas dispersion tube. Connect nitrogen from

cylinder and adjust the flow rate to 150 ml/minute. Start heating the flask and attain boiling within 5 minutes. Continue heating and passing nitrogen for a period of one and a half hours. Disconnect the cylinder containing boric solution and transfer into a 250-ml conical flask. Wash the cylinder and the dispersion tube with distilled water and add the washings to the contents of the conical flask. Titrate immediately against 0.1 N hydrochloric acid. The end point is from green to yellow.



All dimensions in millimetres.

FIG. 1 APPARATUS FOR DETERMINATION OF PROPOXUR CONTENT

A-2.5 Calculation

$$\text{Propoxur content, percent by mass} = \frac{V \times N \times 20.9}{M}$$

where

V = volume of standard hydrochloric acid consumed in the test;

N = normality of hydrochloric acid; and

M = mass of sample, in g, taken for the test.

(AFCD 6)